

Question ID: 2afd3cec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	Hard

Question

After **20%** of the original number of marbles in a group were removed from the group, **360** marbles remained in the group. How many marbles were removed from the group?

Answer

- A. **72**
- B. **90**
- C. **450**
- D. **1,800**

Correct Answer: B

Rationale

Choice B is correct. It's given that **20%** of the original number of marbles were removed from the group. Let *x* represent the original number of marbles in the group. It follows that $\frac{20}{100}x$, or **0.20*x***, marbles were removed from the group. Therefore, *x* — **0.20*x*** marbles remained in the group. It's also given that **360** marbles remained in the group. Thus, *x* — **0.20*x*** = **360**, or **0.80*x*** = **360**. Dividing both sides of this equation by **0.80** yields *x* = **450**. Substituting **450** for *x* in the expression **0.20*x*** yields **0.20(450)**, or **90**. Therefore, **90** marbles were removed from the group.

Choice A is incorrect. This is **20%** of the remaining number of marbles.

Choice C is incorrect. This is the original number of marbles, not the number of marbles that were removed.

Choice D is incorrect and may result from conceptual or calculation errors.